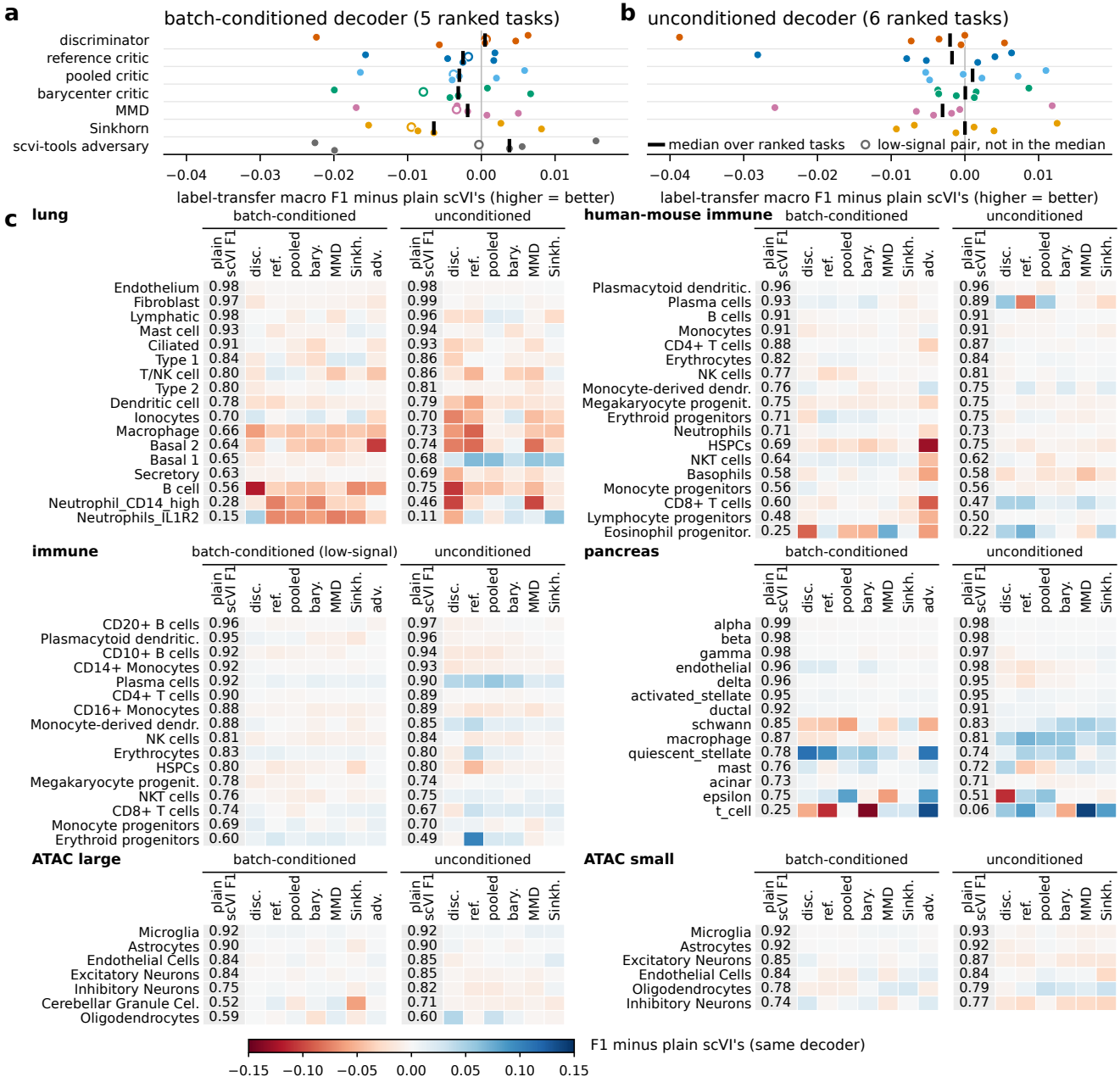

a
batch-conditioned decoder (5 ranked tasks)
discriminator
reference critic
pooled critic
barycenter critic
MMD
Sinkhorn
scvi-tools adversary
−0.04 −0.03 −0.02 −0.01 0.00 0.01
label-transfer macro F1 minus plain scVI's (higher = better)
b
unconditioned decoder (6 ranked tasks)
median over ranked tasks
low-signal pair, not in the median
−0.04 −0.03 −0.02 −0.01 0.00 0.01
label-transfer macro F1 minus plain scVI's (higher = better)
c
lung
batch-conditioned
unconditioned
plain scVI F1 disc. ref. pooled bary. MMD Sinkh. adv.
Endothelium 0.98 0.98
Fibroblast 0.97 0.99
Lymphatic 0.98 0.96
Mast cell 0.93 0.94
Ciliated 0.91 0.93
Type 1 0.84 0.86
T/NK cell 0.80 0.86
Type 2 0.80 0.81
Dendritic cell 0.78 0.79
Ionocytes 0.70 0.70
Macrophage 0.66 0.73
Basal 2 0.64 0.74
Basal 1 0.65 0.68
Secretory 0.63 0.69
B cell 0.56 0.75
Neutrophil_CD14_high 0.28 0.46
Neutrophils_IL1R2 0.15 0.11
human-mouse immune
batch-conditioned
unconditioned
plain scVI F1 disc. ref. pooled bary. MMD Sinkh. adv.
Plasmacytoid dendritic. 0.96 0.96
Plasma cells 0.93 0.89
B cells 0.91 0.91
Monocytes 0.91 0.91
CD4+ T cells 0.88 0.87
Erythrocytes 0.82 0.84
NK cells 0.77 0.81
Monocyte-derived dendr. 0.76 0.75
Megakaryocyte progenit. 0.75 0.75
Erythroid progenitors 0.71 0.75
Neutrophils 0.71 0.73
HSPCs 0.69 0.75
NKT cells 0.64 0.62
Basophils 0.58 0.58
Monocyte progenitors 0.56 0.56
CD8+ T cells 0.60 0.47
Lymphocyte progenitors 0.48 0.50
Eosinophil progenitor. 0.25 0.22
immune
batch-conditioned (low-signal)
unconditioned
plain scVI F1 disc. ref. pooled bary. MMD Sinkh. adv.
CD20+ B cells 0.96 0.97
Plasmacytoid dendritic. 0.95 0.96
CD10+ B cells 0.92 0.94
CD14+ Monocytes 0.92 0.93
Plasma cells 0.92 0.90
CD4+ T cells 0.90 0.89
CD16+ Monocytes 0.88 0.89
Monocyte-derived dendr. 0.88 0.85
NK cells 0.81 0.84
Erythrocytes 0.83 0.80
HSPCs 0.80 0.80
Megakaryocyte progenit. 0.78 0.74
NKT cells 0.76 0.75
CD8+ T cells 0.74 0.67
Monocyte progenitors 0.69 0.70
Erythroid progenitors 0.60 0.49
pancreas
batch-conditioned
unconditioned
plain scVI F1 disc. ref. pooled bary. MMD Sinkh. adv.
alpha 0.99 0.98
beta 0.98 0.98
gamma 0.98 0.97
endothelial 0.96 0.98
delta 0.96 0.95
activated_stellate 0.95 0.95
ductal 0.92 0.91
schwann 0.85 0.83
macrophage 0.87 0.81
quiescent_stellate 0.78 0.74
mast 0.76 0.72
acinar 0.73 0.71
epsilon 0.75 0.51
t_cell 0.25 0.06
ATAC large
batch-conditioned
unconditioned
plain scVI F1 disc. ref. pooled bary. MMD Sinkh. adv.
Microglia 0.92 0.92
Astrocytes 0.90 0.90
Endothelial Cells 0.84 0.85
Excitatory Neurons 0.84 0.85
Inhibitory Neurons 0.75 0.82
Cerebellar Granule Cel. 0.52 0.71
Oligodendrocytes 0.59 0.60
ATAC small
batch-conditioned
unconditioned
plain scVI F1 disc. ref. pooled bary. MMD Sinkh. adv.
Microglia 0.92 0.93
Astrocytes 0.92 0.92
Excitatory Neurons 0.85 0.87
Endothelial Cells 0.84 0.84
Oligodendrocytes 0.78 0.79
Inhibitory Neurons 0.74 0.77
F1 minus plain scVI's (same decoder)
−0.15 −0.10 −0.05 0.00 0.05 0.10 0.15